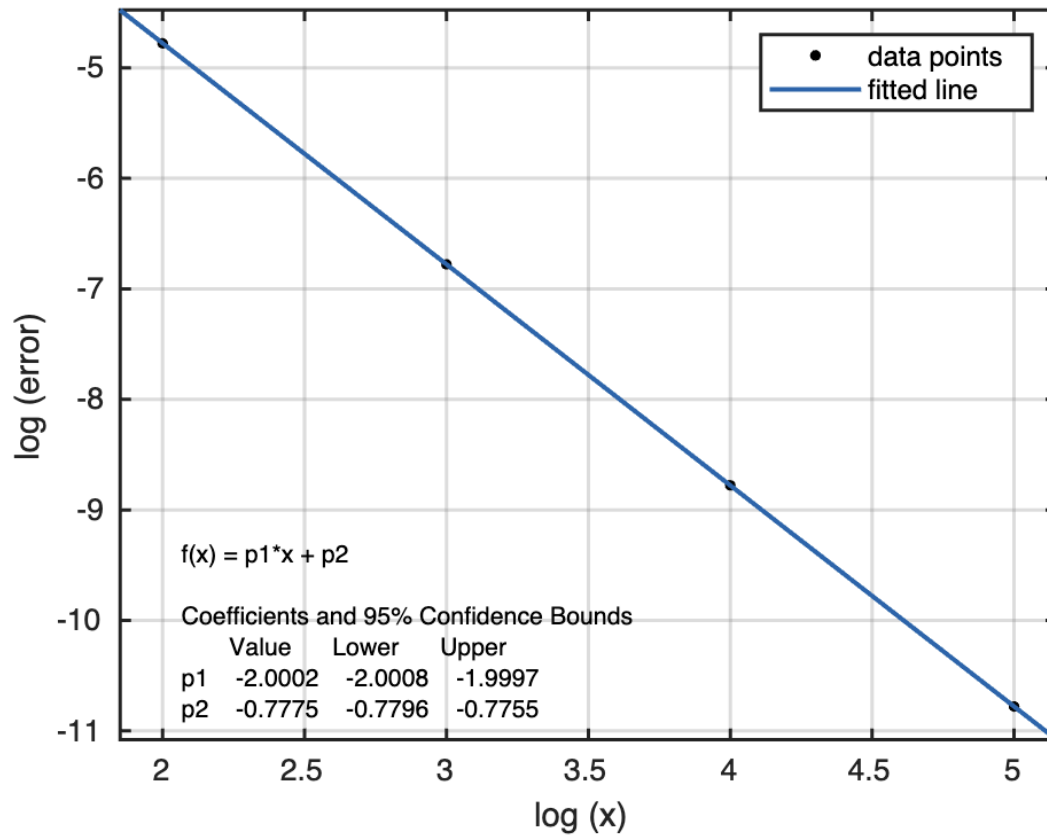


Assignment 2

Question 1

(a) For $n = 100000$, integral = 3.1415926535731553

(b)

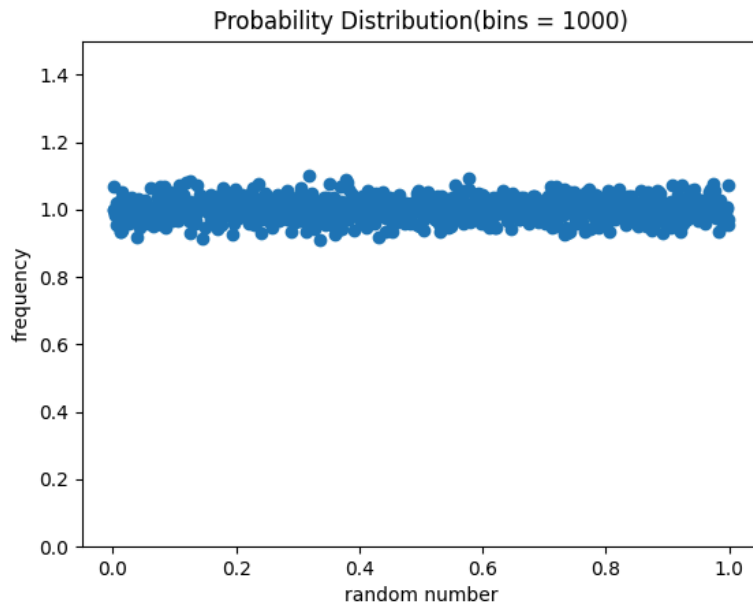


(c) For $n = 100000$, integral = 1.999999998354978

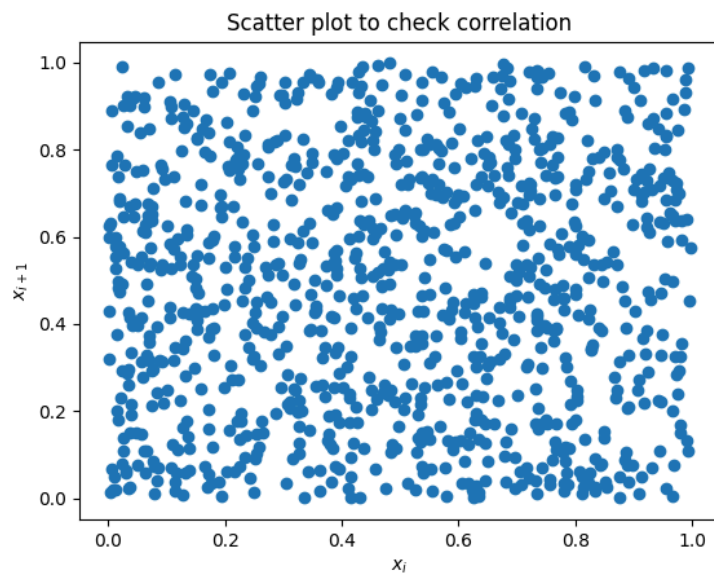
(d) For $n = 100000$, integral = 0.99730020392875796. Yes it matches our expectations since the integral from -3σ to 3σ is around 0.997.

Question 2

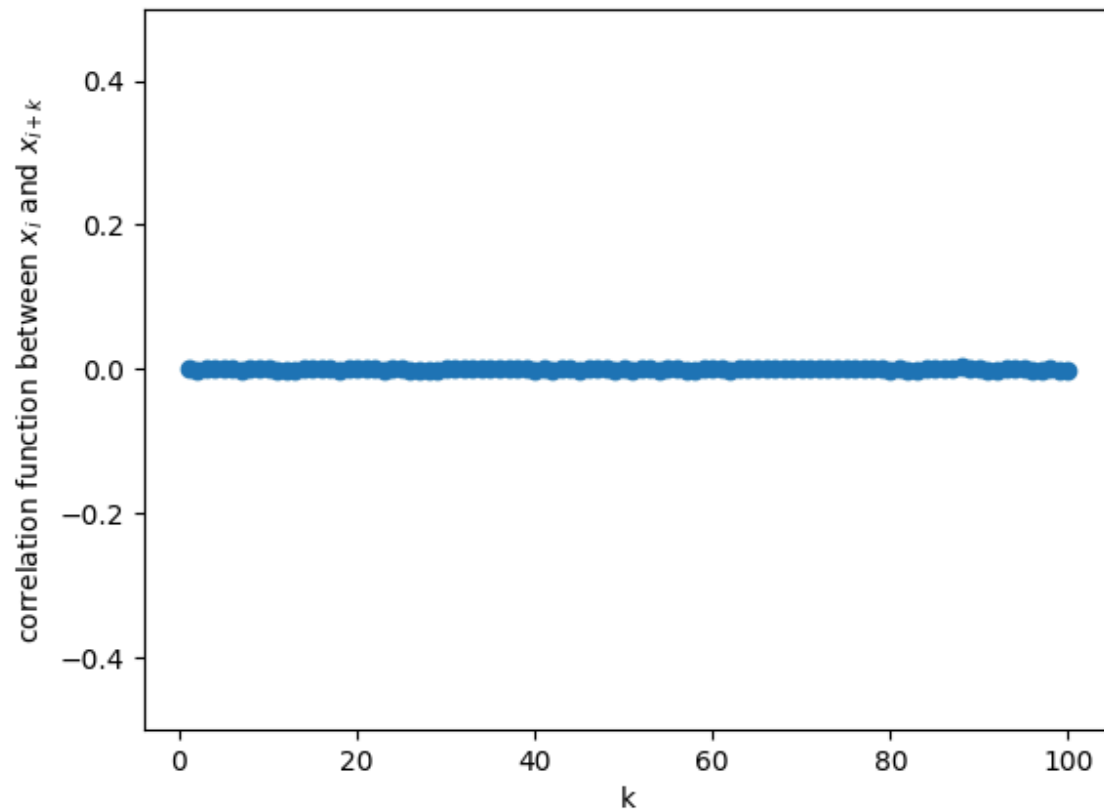
(a)



(b)



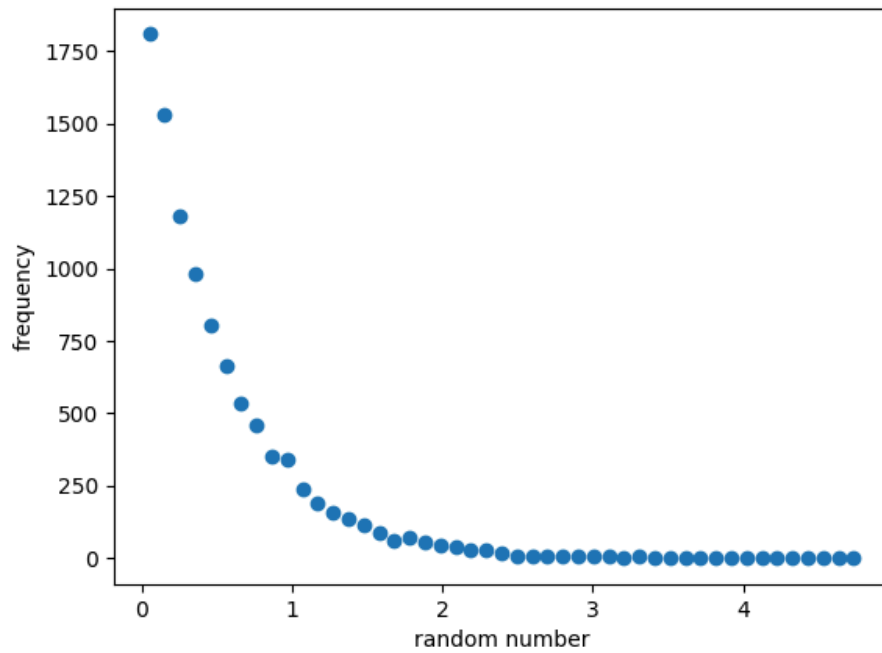
(c) the correlation function for $k = 1$ is
6.5161582730802398E-004



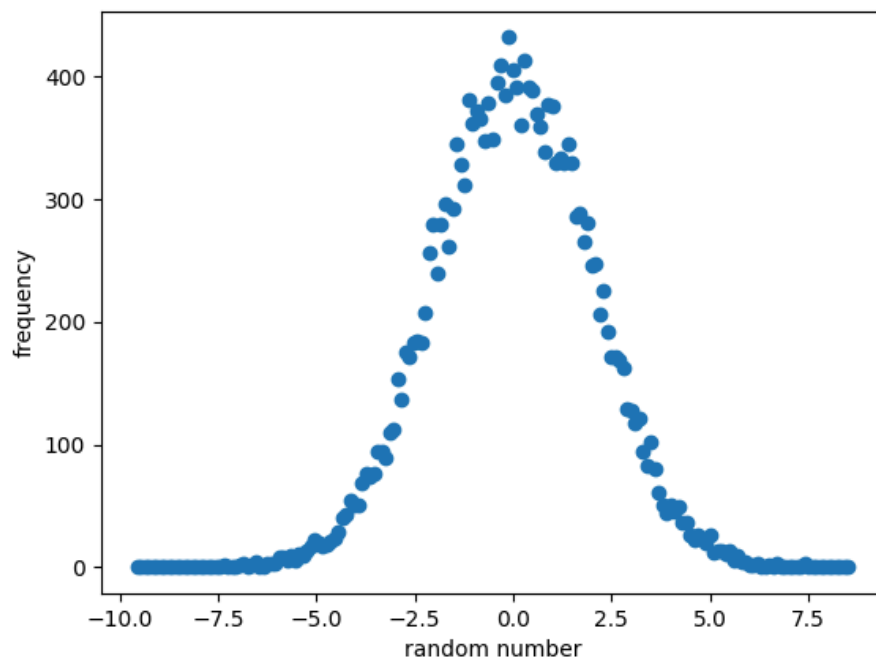
(d) the standard deviation is
0.28861309397857238

Question 3

(a)



(b)



Question 4

(a)	n	Integral	Standard Deviation
	10	6.8421715721010699E-025	6.4910538877687218E-025
	100	1767.0323302951192	1758.1749696048789
	1000	2.0912645489552105E-007	2.0902186491174337E-007
	10000	3.3731289757833014E-003	2.2631296752839754E-003
	100000	14.320527979963746	13.644393147475565
	1000000	8.4514540651373036	4.3276748008816517
	10000000	11.337998263847199	2.8921704334079248
	100000000	11.126721295531452	0.91196008127614181

(b)	n	Integral	Standard Deviation
	10	8.8190053600163356	2.3472763394990803
	100	10.255088082989548	0.76548621725970512
	1000	11.376964634068505	0.25895898171811699
	10000	10.771603567752308	8.0256934478628597E-002
	100000	10.962605732982443	2.5465918487538208E-002
	1000000	10.958435902557170	8.0488807258984729E-003
	10000000	10.964107183692205	2.5467125056310501E-003
	100000000	10.961162359776727	8.0518705817684622E-004